Do AI-measured economic narratives explain why consumer sentiment sits below an economics-only prediction?

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1. Summary of findings

Michigan consumer sentiment is far below what unemployment, inflation and GDP growth predict. An economics-only model fitted on data available at the time predicted 73.4 for October 2026 (preliminary reading 46.3), a gap of −27 points. The average gap since January 2022 is −16 points and −24.5 over the last twelve complete months. **We find no reliable evidence that the tone of the news and social-media coverage we collected explains this gap.** Four results carry that conclusion.

- **The raw association is real but fragile.** Monthly media tone and the gap correlate at +0.49 (50 months, block-bootstrap interval +0.21 to +0.75). A test that respects how persistent both series are gives p = 0.22. The usual HAC test gives p = 0.001, but in our simulation it rejects a true null 17% of the time.
- **The association comes from 2025.** Before February 2025 the correlation is −0.03 (30 months). Once the White House party is controlled for, it falls to +0.02 to +0.04.
- **Media adds nothing to a forecast that already has household and political controls.** Rolling forecast RMSE is 4.57 without media and 4.69 with it; the gain from media is −1.1 (interval −3.7 to +0.0). Media improves a model that has no controls (RMSE 9.95 to 8.50), so the media signal overlaps with what the controls capture.
- **One signal deserves monitoring, not belief.** Changes in tone lead changes in the gap by 1 to 3 months (r = 0.57 to 0.62). With Holm adjustment for seven lags, p is 0.06 to 0.20. The gap does not lead tone.

These are results about what a sample of the Guardian and five Reddit communities published. They say nothing about what survey respondents saw. The study can distinguish tone and topic emphasis in published text. It cannot measure confirmation bias, availability bias in readers, or anchoring, and it does not claim to.

2. Questions, hypotheses and what the data can speak to

The brief poses six questions. We keep findings (what the data show) apart from hypotheses (what would need other data).

Question	Answer from this study	Status
1. How to measure bias in coverage?	Coding scheme fixed before labeling; tone, topic, framing and attribution measured on 20,214 items; validated against 300 human labels. Only tone, topic and relevance passed validation at scale.	Finding (measurement)
2. Effect of bias on Michigan sentiment?	Association only. No causal design is feasible with the data; a randomized framing experiment is proposed (Section 7).	Hypothesis
3. How much do unemployment, inflation, GDP explain?	Out-of-sample R^2 0.45; leaves a −16 point mean gap since 2022. Adding the 5-year price level cuts it to −2 (tentative).	Finding, with caveat
4. Which perceptual mechanisms are	Negativity in published text, topic	Finding (scope)

measurable?	emphasis (availability on the supply side), price-level framing. Not reader-side biases.	
5. Does media improve prediction?	Not beyond economic and household/political controls.	Finding
6. What would contradict media bias?	Listed in Section 8; three of the five are already observed.	Finding

The brief names three competing explanations: pressures that headline indicators miss, selective information or framing, and measurement or model problems. Our evidence favors the first and third as sufficient on their own for much of the gap (the price-level model and survey-mode change) and leaves the second unsupported, not refuted.

3. Economic baseline and the sentiment gap

3.1 Data and timing

The monthly panel runs from January 2010 to October 2026. Sentiment is the Michigan headline index, with current-conditions and expectations sub-indices. Unemployment is the BLS U-3 rate. Inflation is headline PCE, computed as 100 × (PCE price index / its value 12 months earlier − 1) from the BEA price index (FRED PCEPI). Real GDP growth is the quarter-over-quarter annualized rate; quarterly values are held, never interpolated to months. The supplied workbook's PCE column differs from recomputed headline PCE (mean absolute difference 0.54 points), so the model uses the recomputed series and a robustness run uses the workbook column.

Every predictor enters only after its assumed release date (unemployment 10 days after month end, PCE 30 days, GDP 30 days after quarter end). A leakage test confirms every feature was public on or before its prediction date. The 3.7% GDP estimate for the quarter ending September 2026 is released on 30 October, after the cutoff, and is excluded. The October 2025 unemployment reading is missing in the source file and is left missing. The October 2026 sentiment reading is preliminary and is excluded from the media analysis. We do not have vintage data, so revisions to GDP are not tracked; the current vintage differs from the workbook by under 0.05 points over the last ten quarters.

3.2 The gap model

We estimate expanding-window OLS models that refit every month and predict the next month, starting January 2015. The gap is actual minus prediction. Three specifications are compared.

Model	Predictors	RMSE	Out-of-sample R^2	Mean gap 2022-26	Mean gap last 12 mo.
M1 economics only	Unemployment, PCE inflation, real GDP growth	12.60	0.45	−16.2	−24.5
M2 + price level	M1 + 5-year cumulative price level	6.43	0.86	−2.2	−7.6
M3 + inflation expectations	M2 + Michigan 1-year expected inflation	5.87	0.88	−1.4	−6.8

The plain-average benchmark has RMSE 16.97. Before the recent period the M1 model was centered (mean gap +0.7 for 2015-19). A negative gap is unexplained pessimism under the model, not proof of bias. Two cautions apply. First, prediction intervals undercover: nominal 90% intervals contained the outcome in only

62% to 64% of months for M1, because the gap is persistent (lag-1 autocorrelation 0.83). Second, M2's price-level coefficient is learned mostly from the 2021-22 episode, so the large reduction in the gap is a hypothesis about anchoring to past prices, not a finding.

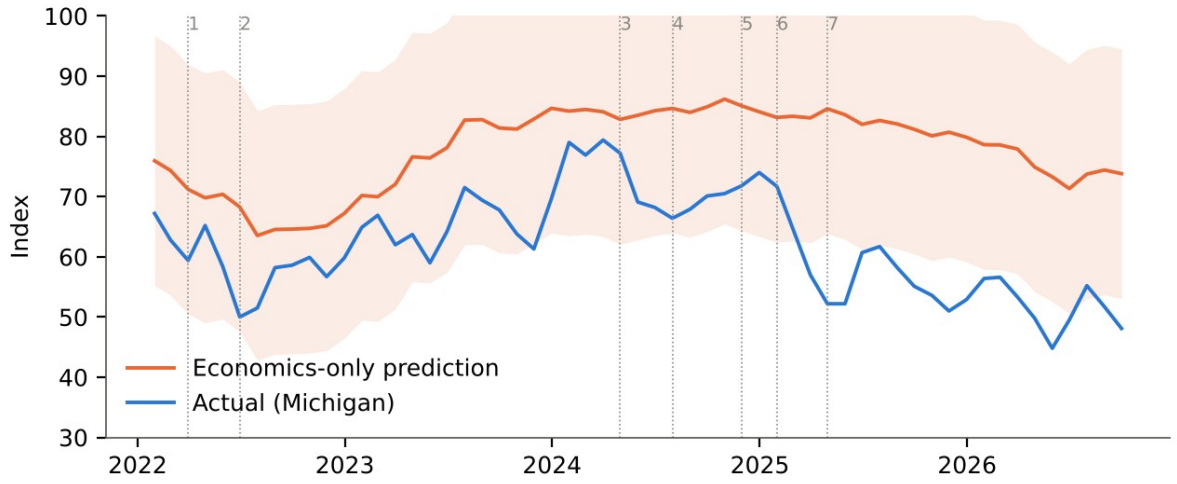


Figure 1. Michigan sentiment against the economics-only prediction. Band: prediction ± 1.645 × out-of-sample RMSE (12.6); it understates uncertainty. Events: 1 Fed begins raising rates; 2 CPI inflation peaks; 3 Michigan web transition begins; 4 survey fully web; 5 election; 6 White House changes party; 7 broad tariffs announced.

3.3 Alternative explanations

- **Prices already high.** Falling inflation does not reverse concern about elevated price levels; M2 adds the cumulative level and shrinks the last-year gap from −24.5 to −7.6.
- **Household experience versus aggregates.** Growth in GDP need not reach household budgets. We could not add real household income, housing affordability or debt-service burdens in time; they are the next controls to add.
- **Survey mode.** Michigan moved to web collection (mixed April to June 2024, fully web from July 2024). The M1 gap was already −11.1 on average in 2022-23, before the transition, so mode cannot be the whole story. Mean gap is −2.4 before April 2024 and −22.2 afterwards, so mode is a live partial explanation and cannot be separated from the 2025 changes.
- **Seasonality and timing.** Calendar-month effects in the gap are absent ($F = 0.29$, $p = 0.99$). Results are robust to dropping the pandemic months and to moving the start of forecasting.
- **Partisanship.** Michigan documents partisan differences in perceptions. We have no respondent-level data, so we cannot split by party, income or age. A White House party indicator is used as a control and as a stand-in for partisan regime.

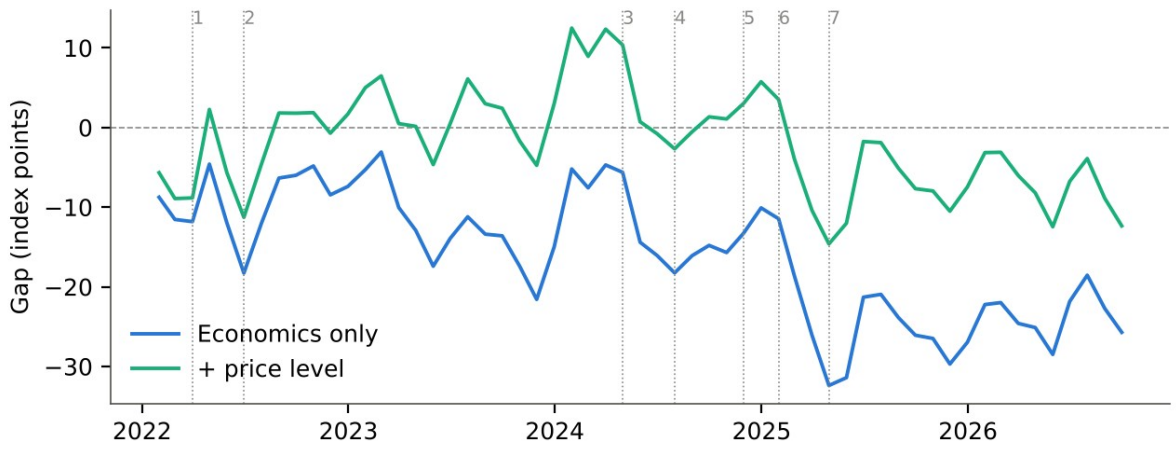


Figure 2. The gap under two models. Adding the price level shrinks the recent gap but the coefficient rests on one episode.

4. Corpus design

The protocol (search terms, sources, inclusion rules) was written to disk before collection and is in the repository. Search terms: economy, inflation, prices, jobs, unemployment, recession, cost of living, tariffs, interest rates, gas prices, grocery, mortgage, wages. Window: January 2022 to October 2026, 57 months.

Source	Unit	Items kept	Notes
The Guardian (US-economy coverage)	Full article	1,992	UK-based outlet with national US desks; full text available
Reddit: r/Economics, r/economy, r/personalfinance, r/politics, r/Conservative	Post (title, 18% with body)	18,222	Weekly archive windows; busy weeks truncated to the earliest posts
GDELT	Headline	135 (dropped)	Rate-limited; excluded from analysis

Filtering took 153,030 collected Reddit and news records to 20,349 after US focus, an economic term in the title, deletion removal, unique URL and near-duplicate collapse; the final analysis corpus is 20,214 items (Guardian plus Reddit). The ten most active Reddit authors account for 10% of posts, and 2% of posts repeat a title within a week. Every month has at least 60 US-economy items from Reddit, but Guardian volume is thin in some months: seven months have fewer than 8 Guardian US-economy items and have no combined tone measure.

What the corpus measures. It measures supply (what was published or posted) and, for Reddit, a sample of discussion. It does not measure exposure. There is no historical local news, no TV, and no X, TikTok or Facebook. Reddit users tend to be younger and more politically engaged than the population, and r/Conservative and r/politics are explicitly partisan. We therefore word every conclusion as being about this corpus.

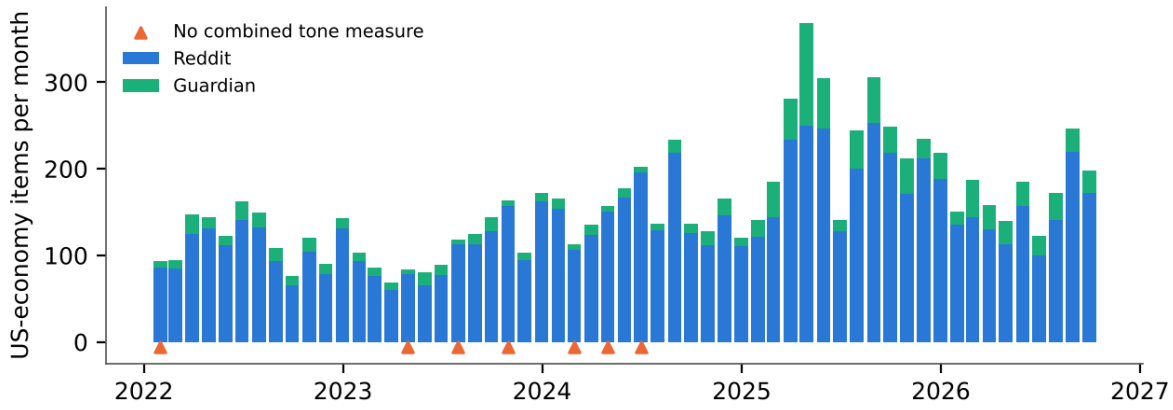


Figure 3. US-economy items per month by source, and months without a combined tone measure.

5. AI measurement and its validation

5.1 Coding scheme and models

The coding scheme (LABELING_GUIDE.md) was fixed before any AI labeling. Each item gets: whether it is about the US economy; topic (prices and inflation, jobs and wages, growth and recession, rates and housing, tariffs and trade, other); tone from −2 to +2; frame (neutral report, risk and uncertainty, cost burden, decline and crisis, resilience and recovery, political blame); blame target; and a confidence rating. Text was truncated to a standfirst plus excerpt (600 characters for Reddit bodies).

An LLM labeled the 300-item validation sample (Claude, via chat, because API credits ran out), and an open model (Qwen2.5-14B-Instruct, AWQ, temperature 0, fixed seed, JSON-schema output) labeled the whole corpus on a rented GPU. The prompt hash, model and cache files are saved. Prompts and settings are in the repository so the open-model labels can be reproduced exactly.

5.2 Human labels and agreement

A stratified sample of 300 items (40 Guardian items and 10 to 19 items from each subreddit per period) was labeled by one annotator (the author) in full and by a second independent rater on a 100-item subset. The brief asks for two annotators on the full sample, which we did not achieve; we report agreement on the subset and treat the author's labels as the reference.

Comparison	Field	n	Cohen's kappa	95% CI
Rater 2 vs rater 1	US economy	100	0.85	0.73 to 0.95
	Topic	58	0.90	0.79 to 0.98
	Tone (weighted)	58	0.90	0.83 to 0.96
Claude vs human	US economy	300	0.88	0.82 to 0.93
	Tone (weighted)	150	0.86	0.80 to 0.90
Open model vs human	US economy	300	0.69	0.60 to 0.76
	Topic	130	0.81	0.74 to 0.89
	Tone (weighted)	130	0.78	0.69 to 0.85
	Frame	130	0.45	0.35 to 0.55
	Blame	130	0.54	0.41 to 0.66

The open model's mean tone is 0.07 points more negative than the human's (−0.75 vs −0.68), within the ±0.30 tolerance set in advance. 100% of its answers were valid JSON. Disagreements with the human cluster at the edges of tone (−1 versus −2) rather than across the neutral line.

Dictionary comparison. A simple negative/positive word list agrees with human tone far less (weighted kappa 0.08, raw agreement 47%; 38% of US-economy items contain none of its words). LLM measurement is therefore worth its cost for tone.

Name and source swap test. Names of Trump, Biden and parties appear in 100 of 300 sampled items. For the open model, swapping them leaves tone unchanged in 91% of swapped items (100% of unchanged controls), and no tone moved by two points or more. Blame labels changed as expected in only 64% of swapped items, so blame is name-sensitive and is not used.

5.3 A validation rule that failed, and how we handled it

Our pre-set rule required kappa of at least 0.60 for every field. Frame (0.45) and blame (0.54) failed, so the original rule failed. We recorded Amendment 3 after seeing the result: only us_econ, topic and tone are used from the open model. This is post hoc and is disclosed here. The earlier amendments are Amendment 1 (Reddit collected in weekly windows to avoid truncation) and Amendment 2 (28 of 300 open-model answers marked not-US-economy but with filled fields were repaired). Framing and political attribution as measured by this model are therefore not results of this study. Factual calibration (whether numeric claims match official data) was not measured; it needs claim extraction and release-time matching that we did not complete.

Independence caveat: the Claude labeler and the author's labels were made in the same project, and the open-model labels were compared with the same human sample, so the validation sample cannot be used to tune anything that is also reported as an out-of-sample result. No tuning was done.

6. Association and prediction

6.1 Monthly media measures

For each month we averaged tone over US-economy items on each platform and combined the two with equal weight, so Reddit's large volume does not swamp the Guardian. A month needs at least 8 items per platform; this gives 50 of 57 months. The salience measure is the share of all items about the US economy; the prices measure is the share of US-economy items on prices and inflation. Median standard error of a monthly mean tone is 0.09 for Reddit and 0.22 for the Guardian, against a month-to-month spread of 0.25 and 0.37, so sampling noise is a modest part of the variation.

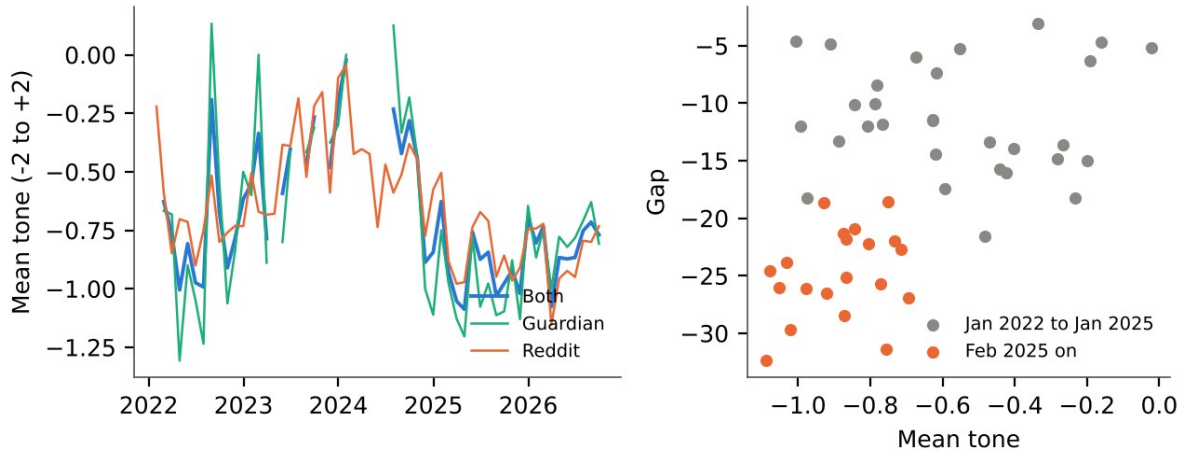


Figure 4. Left: monthly tone by platform. Right: tone against the M1 gap, before and after the White House party change.

6.2 Statistical approach

Both series are persistent (lag-1 autocorrelation 0.84 for the gap, 0.56 for tone), which inflates apparent significance. We simulated this directly: the HAC (Newey-West) test rejects a true null 17% of the time at a nominal 5%. Our main test therefore compares the observed correlation with correlations from phase-randomized surrogates of the media series that preserve its autocorrelation (1,999 draws); its false-positive rate is 6%. We also use 12-month changes (which remove trends and seasonality), moving-block bootstrap intervals, Holm adjustment across 24 specifications, and rolling expanding-window forecasts. Power is limited: with 50 months the minimum detectable correlation at 80% power is about 0.6.

6.3 Does tone track the gap?

Check	Months	Correlation	Surrogate p	HAC p
All months (primary)	50	+0.49	0.22	0.001
12-month changes	33	+0.34	0.28	0.006
Jan 2022 to Jan 2025	30	−0.03	0.95	0.81
Feb 2025 on	20	+0.30	0.44	0.002
Excluding Jan to Jun 2025	44	+0.44	0.31	0.011
Controlling White House party and expected inflation	50	+0.02	0.88	0.81
Controlling White House party only	50	+0.04	0.83	0.68

Across 24 specifications (three gap models, four media measures, level and change), every correlation has the same sign as the primary one. Four have unadjusted surrogate p below 0.05 and none survive Holm adjustment. The strongest is tone against the M2 gap (r = 0.64, p = 0.004, Holm 0.084). Tone from the Guardian alone (r = 0.40) and from Reddit alone (r = 0.52) point the same way. A higher share of items about the US economy goes with a more negative gap (r = −0.46, surrogate p = 0.03 for levels and 0.01 for changes); the share on prices does not (r = 0.28, p = 0.51). The salience result is unadjusted for multiple testing.

6.4 Does media improve prediction?

We compare five models on 26 rolling one-month-ahead forecasts (August 2024 to September 2026). The first 24 months with a media measure are used only for fitting.

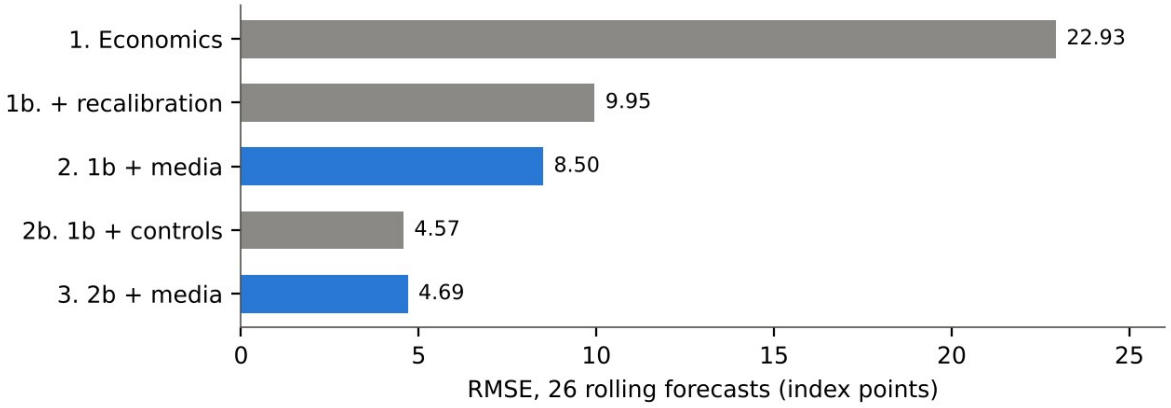


Figure 5. Rolling forecast RMSE. Blue bars include media tone.

Model	RMSE	Compared with	Gain in squared error	95% block-bootstrap CI	DM p (HAC)
1b. Economics, recalibrated	9.95				
2. + media tone	8.50	1b	+26.9	+14.0 to +45.2	0.001
2b. + controls, no media	4.57	1b	+78.2	+40.7 to +130.1	0.000
3. 2b + media tone	4.69	2b	−1.1	−3.7 to +0.0	0.29

Model 3 is compared with 2b, not with the economics-only model, so the difference isolates what media adds. The controls (Michigan expected inflation and the White House party) account for the improvement. Expected inflation comes from the same survey respondents as the sentiment index, so part of its power is mechanical; we flag it as a perception control and not as an independent economic condition. The tone coefficient is near zero until April 2025 and then rises to about +14 (range −0.6 to +14.0 across 26 fits), so it is identified by the early-2025 episode, during which tariffs, the change of administration and the survey becoming web-only coincide. The hold-out is small and persistent, so the Diebold-Mariano p-values are optimistic; the bootstrap interval is the better guide.

6.5 Direction of influence

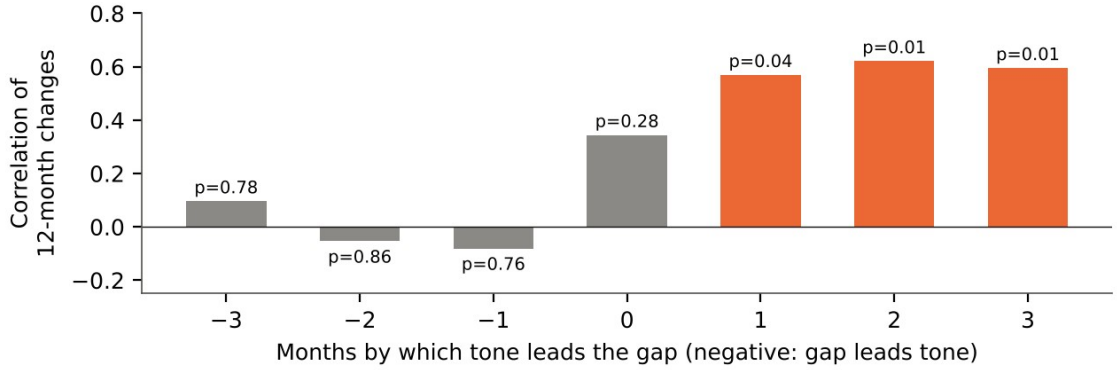


Figure 6. Correlation between 12-month changes in the gap and in tone, with tone leading by k months.

Tone leads the gap by 1 to 3 months (r = 0.57, 0.62, 0.60; surrogate p 0.040, 0.008, 0.012; Holm 0.20, 0.056, 0.069), while the gap does not lead tone (coefficient 0.001, p = 0.73) and last month's tone predicts this month's gap with HAC p = 0.049. That pattern is what a media-influence story predicts. It is also what a

common shock would produce if the media react to events faster than survey responses do. We treat it as an early-warning hypothesis for the post-election update.

7. Causal reasoning

We chose the third design in the brief: observational analysis with a causal-design proposal. No exposure data exist for Michigan respondents.

7.1 Reverse causality and confounding

- **Audience demand.** Pessimistic consumers may seek and post negative content, and Reddit posts are themselves consumer sentiment. Reddit tone is thus partly a sentiment measure, not only a cause. The Guardian is less subject to this; Guardian-only tone shows the same sign (r = 0.40).
- **Common shocks.** Tariffs and price spikes can lower both coverage tone and household mood. The 2025 episode cannot be separated from these.
- **Partisan regime.** A change of White House party shifts partisan perceptions and media framing together, which is why the association vanishes when we control for it.

7.2 Proposed identification: randomized framing experiment

Recruit about 1,200 consenting US adults through an online panel, balanced on party and age, with institutional review approval and a pre-registration. Show all participants the same verified economic facts (the latest PCE inflation, unemployment and GDP figures) under one of three truthful framings: price-level emphasis, inflation-rate emphasis, or a balanced summary with both and a historical comparison. Outcomes are beliefs about price changes, personal-finance expectations, and a short sentiment battery adapted from the Michigan items, measured immediately and after 7 days. With 400 per arm, a standardized effect of 0.2 has about 80% power. Analyze by arm and by party, and correct for the three pairwise tests.

7.3 Transportability

An effect on a panel is not an effect on the national index. Linking it to Michigan needs: (a) exposure estimates of how many respondents encounter each framing and how often; (b) weights that make the panel representative; (c) evidence that effects persist outside a lab and after repeated exposure; and (d) a way to net out partisan sorting into media. Without them the experiment supports only a statement about the mechanism, not a size for the gap.

8. What would contradict a media-bias explanation

Evidence that would contradict it	What we observe
No association before the White House change	Observed: correlation −0.03 over 30 months
Association disappears once partisan regime is controlled	Observed: falls to +0.02 to +0.04
Media tone adds nothing to a model with controls	Observed: gain −1.1, interval includes 0
Gap leads tone as much as tone leads gap	Not observed: gap does not lead tone
Framing experiment finds no belief change under alternative truthful framings	Not yet tested

Two things are not contradicted: the possibility that media influence sentiment through channels we did not measure (local news, television, social video) and the possibility of an effect too small to detect with 50 months. Absence of evidence here is limited by power.

9. Limitations and ethics

- **Corpus.** One national outlet (UK-based), five Reddit communities, no local news, TV or other social platforms. Reddit weeks with many posts are truncated to the earliest posts. Reddit users are not representative.
- **Measurement.** One main human annotator; a second on 100 items. The open model's frame and blame labels failed validation, and Amendment 3 was post hoc. Factual calibration was not measured.
- **Model.** The economics-only model is a simple linear one. Prediction intervals understate uncertainty. The price-level result rests on one episode. We lack real-time vintages for GDP and revisions.
- **Inference.** Few independent observations (the gap is one persistent series). Power is low. The association and forecast results are driven by a small number of months in 2025.
- **Survey mode.** The shift to web collection overlaps the period of interest and could not be isolated.
- **Ethics.** All data are public or licensed (Guardian Open Platform under its terms; public Reddit archive posts). We store usernames only inside raw archived records, do not report individuals, and make no claims about private persons. The proposed experiment needs institutional review before any human-subject data collection; none was collected. We do not equate criticism of an administration with bias, and tone labels describe wording, not accuracy.

10. Cutoff, prospective update, and one practical implication

The analysis uses information available by 13 October 2026. We do not attribute any change to election proximity merely because it occurs during the campaign. After the 3 November election, the update plan is: refresh Michigan files after the final October release, collect Guardian and Reddit items for October and November, rerun the notebook cells for labeling and Part 5, and compare October and November sentiment with the predictions frozen at the cutoff before any refit. The dashboard rebuilds from the saved tables.

Implication for forecasting and risk management. When nowcasting Michigan sentiment, recalibrate the economics model and add the price level and inflation expectations; media tone is not worth the pipeline cost as a forecasting input. Treat tone as a monitor: a 1 to 3 month lead of tone changes over gap changes is a reason to look, not to trade. Widen the model's intervals, since the 90% band covered 64% of months, and expect the gap to persist (autocorrelation 0.83). For positions sensitive to consumer confidence, the main risk is regime change in the sentiment-to-economy relationship, not the news cycle.

Appendix: reproducibility

The repository contains one Jupyter notebook (final_project.ipynb) that rebuilds every table here from cached downloads (data/cache with manifest.json; nothing is downloaded twice), a Colab notebook that runs the open model (colab_open_model_labeler.ipynb), the labeling guide (LABELING_GUIDE.md), the saved prompt and its hash, frozen label sheets and keys, and requirements.txt. Data provenance: Michigan survey files, BLS, BEA, FRED, Guardian Open Platform, and the Reddit archive. Protocol amendments are recorded in the protocol file and in Section 5.3. The dashboard rebuilds from the day5 CSV outputs.